

**After Economy:
Constraint, Price, and
the End of the Fiscal World**

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May 2026

Abstract

Price is a late institutional interface, a form of commensuration that appears after value has already become consequential. We trace a sequence that runs through constraint, flow, cost, allocation, binding relation, valuation, commensuration, price, money, market, capital, and financial abstraction. Physical systems supply gradients and dissipation, to which economic language applies only by analogy. Cells allocate energy, metabolites, membrane gradients, and repair capacity under viability constraints and incur costs and tradeoffs without prices. Organisms budget across energy, attention, immune response, movement, and reproduction, and sensorimotor coupling, such as a catcher intercepting a ball, is transaction without exchange. Human institutions stabilize these older structures: gifts create debt without money, blood compensation creates price without markets, and states fiscalize grain, land, labor, and debt before capitalism. Capitalism subordinates value, money, markets, and price to accumulation, and financialization prices claims on future income, risk, collateral, and state rescue. “After economy” is defensible only in a narrow sense, as the loss by monetary price of its monopoly over the public interpretation of value; computation, platforms, ecological accounting, and public provisioning shift allocation into compute, attention, energy, access, and governance without removing it. A stylized six-sector allocation model illustrates the structure: undistorted prices reproduce the constraint optimum by construction, while a markup of 3 on finance raises its budget share from 0.02 to 0.26 and lowers total viability by 8.08 percent. These figures describe the model’s geometry and do not measure any real economy.

1. Introduction

Modern institutions often treat price as the public test of reality. A forest becomes real to an institution when it enters a balance sheet, a carbon market, a concession, a tax schedule, or a damages claim. A person’s time becomes real as wage time, billable time, platform time, or debt-service time. Attention becomes real when it is sold at auction, and risk when it is priced as a premium, spread, derivative, or reserve requirement.

This arrangement is historical. Price is a technique for stabilizing commensuration: it makes heterogeneous things comparable under a unit and an institution, and it can attach to bread, labor, rent, injury, land, risk, carbon, reputation, and attention. The attachment follows a value frame that has already made the thing socially, biologically, legally, or politically consequential. Blood matters before blood compensation, labor before wages, land before rent, attention before advertising auctions, and carbon before carbon credits.

We call the relevant threshold *economyicity*. A relation becomes economy-like when constrained flows are recurrently allocated under a value frame and a stabilizing mechanism, which may be metabolism, habit, kinship, ritual, law, fiscal administration, money, platform code, or ecological governance. A river has constrained flow without economic organization. A cell dividing its proteome between growth and repair has economy-like organization, because the alternative pathways matter for its viability. A gift relation is economic because obligation, prestige, delay,

and reciprocity stabilize transfers over time. A state has an economy once land, grain, labor, and population become legible to accounts and extraction.

Economy therefore begins after bare constraint and before money and markets, along the chain

constraint → flow → cost → allocation → bind → valuation → commensuration → price.

Money generalizes some prices, markets produce and discipline some prices, capital reorganizes price around accumulation, and finance prices claims on future conditions. Monetary societies tend to treat one interface in this chain as the truth of the whole. Sections 2 to 10 follow the chain from physical constraint to financial and computational allocation, and section 11 illustrates its central structural claim with a small allocation model.

2. Constraint and flow

Constraint is more general than economy. A constraint reduces the set of possible states or paths, and constraints may be physical, chemical, biological, social, legal, symbolic, or computational. A membrane constrains molecular movement, a rule constrains access, a taboo constrains action, a price constrains demand, an algorithm constrains visibility, and a carbon budget constrains permissible combustion.

Flow is the counterpart of constraint. Matter, energy, information, affect, obligation, signs, money, and risk move through channels whose structure selects what can pass. Deacon's (2011) account of constraint, the dissipative structures of Prigogine and Stengers (1984), and Schrödinger's (1944) treatment of life as an order-maintaining process anchor the social concept of economy in physics. They also set a limit. Dissipation alone does not constitute an economy: a falling rock does not allocate among alternatives that carry value. Physical systems supply gradients, channels, barriers, work, and loss. Economicity appears when constrained flow is distributed among alternative pathways that matter for persistence and reproduction, or for rank, power, legitimacy, and value.

Biology is the first level at which the economic analogy becomes analytically useful. Living systems maintain themselves by regulating flows under constraint. A cell is bounded, selective, and temporally organized, and it must divide resources among growth, repair, replication, movement, defense, dormancy, and death. A bacterium under nutrient stress shifts gene expression, metabolism, and repair in ways that can be modeled as resource allocation (Molenaar et al. 2009; Scott et al. 2010). The analogy invites anthropomorphism and must be used with care: cells allocate through regulation, without calculation, property, bargaining, or settlement.

Cost enters at this level in a restricted sense. Physical cost can mean energy dissipated or exergy lost; biological cost means expenditure relative to viability, repair, replication, or survival; social cost means a burden under a norm or institution; economic cost means alternatives foregone under a value regime. The word is the same across these levels, and its reference changes.

3. Cost in cells and organisms

ATP is often called the cell's energy currency, and the phrase holds only within strict limits. ATP couples otherwise unfavorable biochemical reactions to cellular work; it is generated, hydrolyzed, regenerated, and routed through pathways. It lacks the institutional features of money, namely a unit of account, social payment, legal settlement, durable store of value, property transfer, and public price. It resembles a currency only as an energy carrier.

The cellular case remains informative. Protein synthesis, ion transport, motility, DNA repair, membrane maintenance, and replication all impose costs. A cell can invest in growth at the expense of stress resistance, or in repair at the expense of replication speed. Resource-allocation models in cell biology are useful because the cell faces constrained alternatives, so cost is present where price is absent.

Organisms extend the same structure across more registers. Glucose, oxygen, fat stores, immune activation, sleep, pain, reproduction, risk, and attention are all distributed under constraint. Fever consumes energy, immune responses can damage tissue, reproduction competes with repair, vigilance competes with rest, and movement exposes the organism to danger while securing food or mates. Physiology budgets before any reflective calculation.

Read as analogies, these cases block a common reduction. Cost does not imply money, and allocation does not imply market exchange: a system can expend, conserve, defer, repair, and sacrifice without pricing anything. Price is a later public technique for making some of these burdens commensurable, and section 11 illustrates this claim in a small allocation model.

4. Sensorimotor transaction

A ball and a catcher illustrate transaction without exchange. A ball in flight constrains possible futures through gravity, spin, speed, trajectory, distance, and time to contact. The catcher enters this field: vision tracks the motion, posture adjusts, the body moves to reduce error, the hand opens, and the impact is absorbed. Ball and catcher form a temporary dynamical coupling in which the future state of each is constrained by the relation.

In this broad sense a transaction is a state-changing coupling between systems. Exchange is narrower: it requires a recognized reciprocal transfer, and social exchange requires conventions or agents able to treat the transfer as a transfer. Monetary exchange adds property, price, settlement, and accounts, and a capitalist transaction adds the advance of value for its augmentation.

Gibson's (1979) concept of affordance explains the example. The ball affords catching only relative to a body with specific capacities, position, skill, attention, and timing. Uexküll's (1934) *Umwelt* makes the same point from another direction: the world that matters to an organism is structured by its sensorimotor form. Enactive cognition (Varela et al. 1991) and active-inference models (Friston 2010) describe the coupling in detail, with the organism acting to maintain a workable relation to the constraints it meets.

The catch is a dynamical transaction under a task-relative value frame. Interception matters to the system, energy and time are spent, and the catch succeeds or fails. The example shows that monetary transaction is a derived case, and purchase one late species of transaction.

5. Gift and obligation

The gift is a social form of allocation without monetary price. Mauss's (1925) triad of obligations, to give, to receive, and to return, describes a system in which objects move but the object is never the whole event: prestige, honor, delay, dependency, alliance, and memory move with it. A gift received creates an asymmetry that must be borne until it is answered, transformed, refused, or converted into subordination.

The gift thus produces debt without money and value without price. The return gift may be measured loosely, ritually, competitively, or symbolically, and it may have no scalar equivalent. The debt may be experienced as gratitude, shame, obligation, rank, or danger. Bourdieu's (1972, 1979) analysis of symbolic capital guards against treating the gift as a sentimental alternative to exchange: generosity can dominate, delay can discipline, and recognition can bind more tightly than contract.

Polanyi's (1944) concept of embeddedness remains useful because it undoes the assumed naturalness of market exchange. Provisioning, reciprocity, redistribution, and householding operate before pricing enters them. Sahlins (1972) on Stone Age economics and Graeber (2011) on debt extend the same correction, although both require grounding in local history. Money is one way to settle an obligation, and obligation is both older and wider than money.

6. Blood compensation

Blood compensation is the clearest case of price outside markets. Wergild, blood money, fines, legal damages, penance, and compensation schedules each convert injury, death, guilt, or vengeance into a compensable form. The payment is a legal and ritual commensuration that contains violence and settles rank, responsibility, and closure.

Its functions can be stated separately. Commensuration turns harm into something payable or performable. Containment prevents revenge from extending indefinitely. Hierarchy appears when compensation varies with status, kin position, gender, or political order. Sovereignty appears when the authority to set compensation, accept settlement, or forbid revenge passes from kin groups to courts, rulers, churches, or states (Hyams 2003).

Nietzsche's (1887) genealogy of creditor, debtor, memory, pain, and punishment belongs to this discussion as philosophy, with legal history left to legal-historical sources. Girard (1972) and Bataille (1949) are relevant because sacrifice, expenditure, and violence lie near the boundary between compensation and order. Legal anthropology supplies the empirical check: a blood price combines kinship, feud, masculinity, status, cosmology, penalty, and authority in a specific arrangement, and the number is one surface of the settlement.

The claim can be stated without overreach. Price can exist without a market, and value without a price. A price in the full sense requires a regime of valuation and commensuration behind it; without such a regime the word is a metaphor.

7. Fiscal administration before capitalism

States inherit provisioning, exchange, and obligation and change the scale and medium in which they are made legible. Fiscalization begins when land, grain, labor, animals, debts, persons, and days become recordable claims. Mesopotamian tablets are central evidence, recording accounting, rations, silver and barley equivalences, temple and palace administration, labor obligations, and debts long before capitalism (Nissen et al. 1993). Writing in this setting served as an administrative technology for making flows durable, comparable, and enforceable.

Scott's (2017) account of early grain states supplies the ecological and administrative dimension: grain is taxable because it is visible, divisible, storable, measurable, and tied to sedentary fields. Goody (1986) and Schmandt-Besserat (1992) supply the dimension of media, showing how lists, tablets, tokens, seals, and writing change what can be counted and transmitted. Hudson's (2002) work on ancient debt and clean slates supplies the political dimension: creditor claims can threaten social reproduction and require sovereign cancellation.

Economic activity has no single origin. The modern economy as a bounded macro-object is much later and depends on statistics, national accounting, and policy expertise, as Mitchell (1998) argues for the twentieth century. Mesopotamia contributed to the invention of fiscal reality, the conversion of heterogeneous life into accounts that an institution can store, compare, command, and settle.

Fiscalization is an ancestor of later monetary societies because it trained institutions to treat the world as an accountable surface. Taxes, rations, debt ledgers, censuses, tribute, land surveys, and labor obligations each turn persons and things into units of obligation. The ledger preceded the market as a governing interface.

8. Money, markets, and capital

Money generalizes commensuration. Simmel's (1900) account of money as abstraction and social distance remains one of the clearest descriptions of this change: money loosens things from local relations by giving them a common form. Marx (1867) describes the further transformation in which the money-form emerges as the universal equivalent and becomes the medium through which commodities, labor-power, and capital circulate. Weber (1905) adds the discipline of calculability in bookkeeping, bureaucracy, vocation, methodical conduct, and rationalized enterprise.

Markets are older and broader than capitalism. Braudel's (1979) distinction between material life, markets, and capitalism prevents the reduction of all exchange to capital. Markets coordinate through repeated exchange under norms, property, and price; capitalism reorganizes market forms around accumulation. Marx's formula $M - C - M'$ captures the shift, in which money is advanced in order to return as more money and labor-power, land, tools, debts, and time are drawn into the circuit.

The chartered corporation intensified the circuit by joining accounting, monopoly, violence, and sovereignty. The English East India Company and the VOC combined corporate personality, state authorization, armed force, monopoly privilege, territorial administration, and fiscal extraction (Stern 2011). The plantation ledger applied the same logic more directly, recording enslaved people as assets, labor capacity, depreciation, discipline, insurance risk, and future output.

Capitalism inherits value, price, money, and markets, subordinates them to accumulation, and treats capitalized price as the privileged public measure of value. Whatever cannot enter that form is externalized, enclosed, moralized, subsidized, or destroyed.

9. Financialization and computational allocation

Financialization extends fiscalization into the future. Grain, labor, land, and tax were the earlier objects of legibility; future income, volatility, risk, debt service, collateral, liquidity, and state rescue are the objects of finance. Krippner's (2011) definition of financialization as a pattern of accumulation in which profits accrue primarily through financial channels remains useful when kept material (see also Epstein 2005; Arrighi 1994). Finance remains anchored in production, rent, households, public guarantees, ecological extraction, and future solvency, because it prices claims on them.

Financial abstraction is recursive because prices become inputs to further prices. A mortgage becomes a security, the security becomes collateral, and the collateral supports leverage. Risk models price the instrument, and the instrument's price feeds back into risk and liquidity models. The result is enforceable claims layered on enforceable claims, held together by law, accounting, counterparties, central banks, and the expectation that someone will absorb the losses when the chain fails.

Computation changes the form of allocation without removing the problem of power. Hayek's (1945) argument that prices carry dispersed knowledge and Mises's (1920) calculation argument remain serious. Beer's (1972) cybernetics, the computational planning proposed by Cockshott and Cottrell (1993), platform logistics, and contemporary AI all reopen the technical question. None of them settles preference formation, tacit knowledge, legitimacy, distributional conflict, corruption, ecological constraint, or the choice of objective function.

Platforms already allocate without ordinary market prices at many interfaces (Srnicek 2017; Zuboff 2019). Feeds allocate attention, a resource whose scarcity Simon (1971) identified early, and search allocates visibility; credit models allocate trust, hiring systems access to work, and insurance models risk categories. Cloud systems allocate compute, and recommender systems allocate cultural exposure. These mechanisms often coexist with prices, auctions, subscriptions, and rents. A move beyond price may therefore consist of more calculation while leaving domination intact.

10. Ecological limits and post-scarcity

Post-scarcity is coherent only when its object is specified. Digital copying can approach abundance because the marginal cost of reproduction is low, and symbolic production has become cheap in

some domains. Open-source software and peer-produced knowledge show that valuable work can be organized partly outside direct price (Benkler 2006), although it still depends on hidden labor, firms, universities, states, protocols, energy, devices, and time. Public provisioning can decommodify some necessities, and commons can be governed durably outside both state and market (Ostrom 1990). None of these facts abolishes material constraint.

Ecological economics sets the hard boundary. Georgescu-Roegen's (1971) entropy argument and Daly's (1977) steady-state economics treat economic systems as throughput systems embedded in biophysical limits. Energy, carbon, water, metals, soil, biodiversity, heat, waste, land, and repair remain allocation problems. Carbon budgets become price-like only when an institution makes them tradable or taxable; in themselves they are accounts of constraint.

Banks's (1994) description of money as a sign of poverty serves as a test. In a monetary society, money signifies access. In a post-scarcity regime in the full sense, necessities would be decommodified, with food, shelter, care, education, communication, transport, and cultural participation guaranteed independently of individual purchasing power. Money would then mark optional scarcity: positional goods, unique places, prestige, danger, or luxuries. If money disappears first from luxury while survival remains priced, the result is abundance for an oligarchy.

The phrase "after economy" therefore names a choice between two paths. One weakens the monopoly of price through commons, public provisioning, democratic planning, ecological constraint, and machine coordination accountable to political judgment. The other completes the reach of economy, with every gesture scored, every stream of attention auctioned, every ecological limit financialized, every access right tiered, and every model optimized for an owner. The same technologies make both paths more feasible.

11. A stylized allocation model

The claim that price is an interface layered on allocation, and not its substrate, can be illustrated on a small model, although not measured by it. The model exhibits the direction of a mechanism and is built so that the mechanism appears as the argument expects. A system divides a fixed budget $B = 6$ across six sectors (subsistence, infrastructure, care, knowledge, ecology, and finance), each with concave viability return $V_i(x) = w_i \log(1 + x/s_i)$ and common scale $s_i = 0.5$. Finance has a low true weight, stipulated at 0.30, on the assumption that it coordinates claims and does not itself sustain the system. Three coordinators allocate the same budget. The constraint coordinator maximizes total viability under the budget alone, with no prices, in the manner of a cell dividing energy among repair and maintenance. A faithful price coordinator sets every price to 1. A financialized coordinator applies a speculative markup d to the price of finance, $p = 1 + d$, and maximizes price-weighted return $\sum_i p_i V_i$.

Each allocation has a closed form. The first-order conditions give $x_i = p_i w_i / \lambda - s_i$ with $\lambda = \sum_i p_i w_i / (B + \sum_i s_i)$ whenever every sector receives a positive share, which holds for every markup examined. The simulation computes the allocations by bisection on λ and checks them against this closed form, to within 9×10^{-16} .

The constraint optimum reaches a total viability of 4.97 and assigns 0.02 of the budget to finance. The faithful price coordinator reproduces that allocation, with a divergence from the optimum of 0.0. This identity holds by construction: with all prices equal to 1 the faithful coordinator maximizes the same objective as the constraint coordinator, so the two must return the same allocation. The model illustrates a definitional point, that an interface without distortion is transparent to the allocation beneath it.

A markup on finance removes that transparency, and this result is also a consequence of the construction. At a markup of 3 the financialized coordinator raises finance from 0.02 to 0.26 of the budget and loses 8.08 percent of total viability relative to the optimum. The loss rises monotonically and faster than linearly with the markup: at a markup of 8 finance takes more than half the budget and viability falls by 25.1 percent. That the over-allocated sector wastes viability follows from its stipulated low weight; a markup on a high-weight sector would not appear as waste in the same way. These numbers describe the model's geometry and measure no actual economy. The construction shows the mechanism and its direction and is silent on the magnitude, or the existence, of any real misallocation.

The model also sets aside a problem that the argument of section 9 takes seriously. The constraint optimum is computed by an allocator with full information about every sector's true viability return, which is the omniscient planner that Hayek (1945) and Mises (1920) argue no central authority can be. The model does not contest their argument. It assumes the full-information optimum as a reference point to show what an undistorted objective would yield, and it says nothing about whether any real planner, market, or computation could discover that optimum from dispersed and tacit knowledge. What the geometry shows is the gap between a distorted and an undistorted objective, under the generous assumption that the undistorted objective is available. Allocation under constraint is held fixed in the model; the recursive markup of financialization represents the way an interface can detach from the allocation it was meant to track, at a cost in viability that the prices themselves do not display.

12. Conclusion

Price can lose its authority while constraint, cost, allocation, obligation, scarcity, risk, and power remain. The defensible claim is narrow: monetary price is losing its monopoly as the public interface of value, and other allocation regimes are becoming explicit.

The preceding sections separate features that monetary societies treat as a single bundle. Cells incur cost without price. Sensorimotor coupling is transaction without exchange, the gift creates debt without money, and blood compensation sets price without a market. Fiscal states made the world legible before capitalism. Capital subordinates price to accumulation, and finance prices claims on future conditions and then prices those prices. Computation allocates through prediction, ranking, and access, and ecology sets the limit of every claim to abundance.

The question raised by the phrase "after economy" is therefore who controls allocation. If price no longer appears as the measure of what exists, other institutions will determine what matters and

what circulates, what is scarce, and who bears the cost of scarcity, and which institutions these will be remains undecided.

Reproducibility

The allocation model is implemented in `simulation/allocation.py`. Running `uv run run_all.py` in `simulation/` computes the three allocations, the markup sweep, and the closed-form check, and writes all values to `simulation/output/results.json` and the figures to `simulation/output/figures/`.

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